

QUESTION 4

Compare Rule-Based and Machine Learning-Based Email Spam Detection Techniques. Analyze Their Effectiveness in Handling Evolving Spam Patterns and Minimizing False Positives.

1. Introduction

Email is one of the most widely used methods of communication in organizations. However, email systems are frequently targeted by **spam, phishing, malware, and fraudulent messages**.

To identify unwanted emails, two major techniques are commonly used:

1. **Rule-Based Spam Detection**
2. **Machine Learning-Based Spam Detection**

Rule-based systems depend on predefined rules created by administrators, whereas machine learning systems learn patterns from previously classified email data.

Both techniques have advantages and limitations. The major challenge is to detect **new and continuously changing spam patterns** while minimizing **false positives**, where legitimate emails are incorrectly classified as spam.

2. Rule-Based Email Spam Detection

Rule-based spam detection uses predefined conditions to identify suspicious emails.

The administrator creates rules based on characteristics such as:

- Sender address
- IP address
- Domain reputation
- Subject keywords
- Email content
- URLs
- Attachments
- Header information

For example:

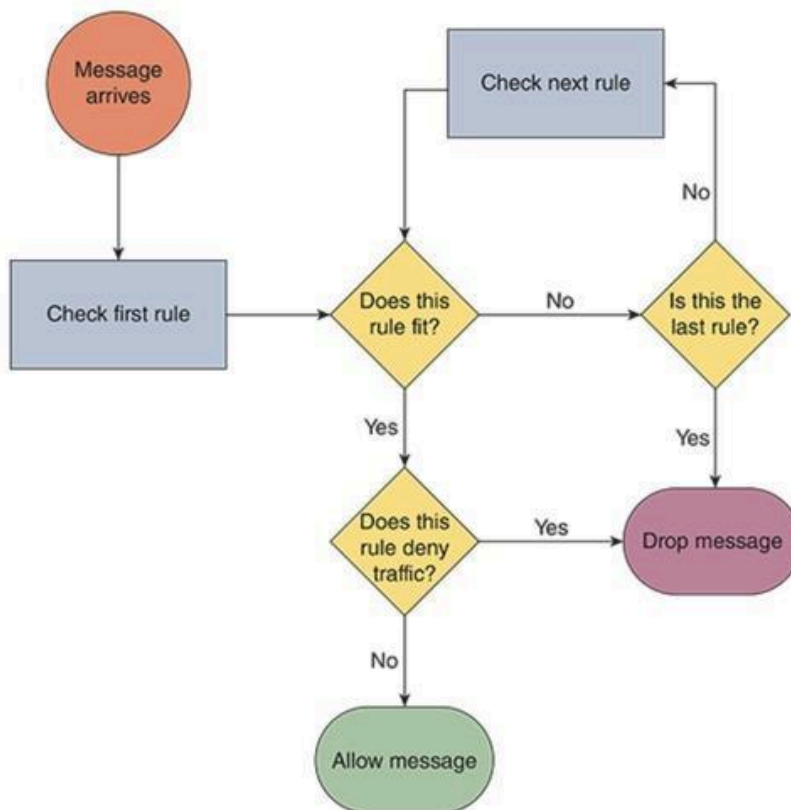
IF sender is blacklisted
THEN classify as SPAM

Another rule could be:

IF email contains suspicious URL
AND contains "urgent"
THEN increase spam score

Traditional filtering systems commonly combine several filters, such as blacklist/DNSBL checks and content or rule-based scanning.

Picture 1 – Rule-Based Spam Filtering



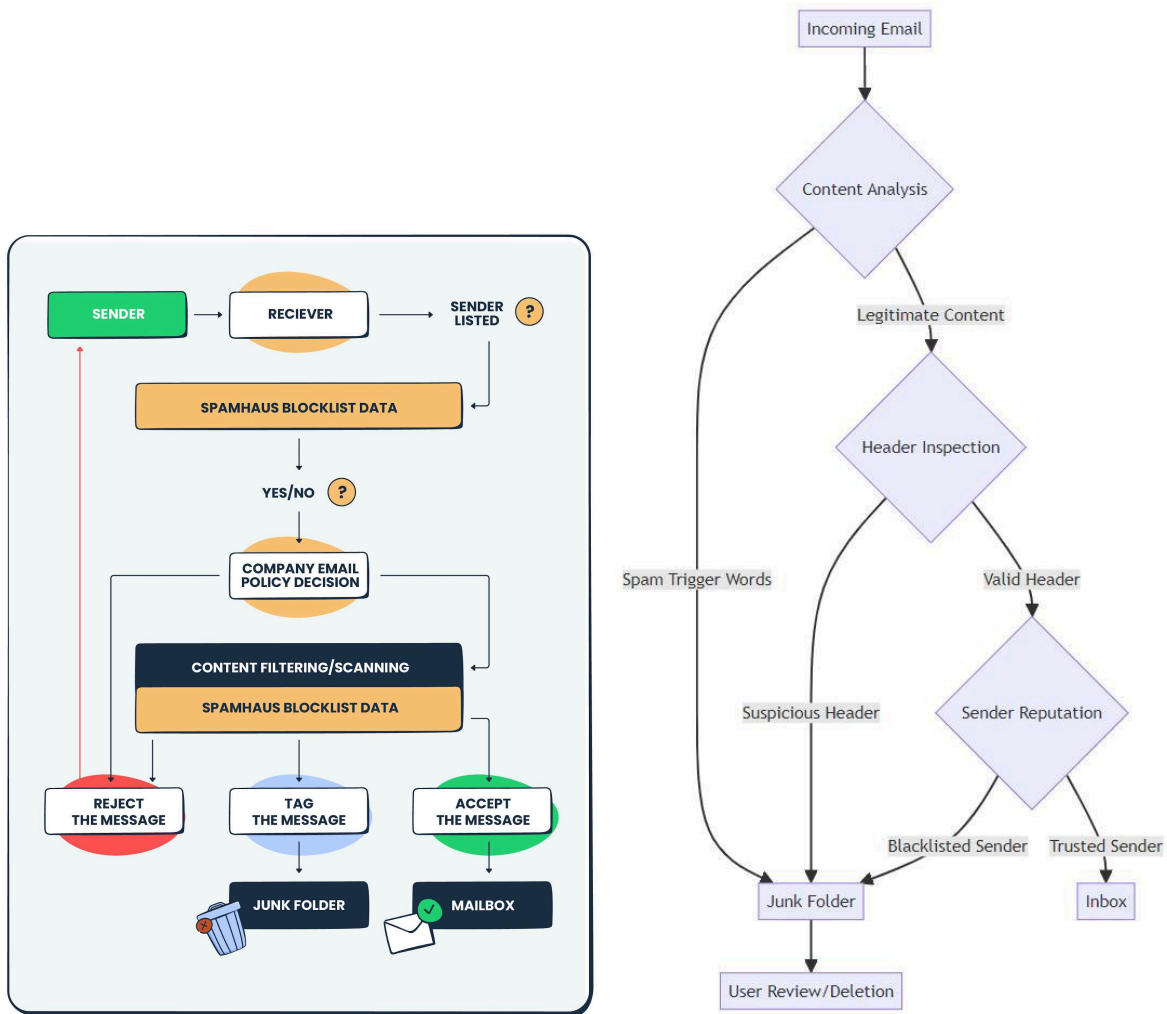


Figure 1: Rule-based email spam filtering process

Working

Incoming Email



Check Sender



Check Blacklist



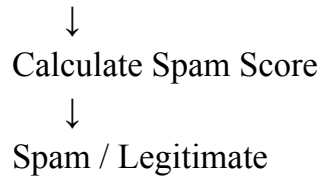
Check Keywords



Check URL



Check Attachment



The main advantage is that the decision is easy to understand because the system can identify exactly which rule caused the email to be classified as spam.

3. Advantages of Rule-Based Detection

3.1 Simple Implementation

Rule-based systems are relatively simple to design and configure.

For example:

IF sender = blacklisted
THEN reject email

The administrator can easily understand the rule.

3.2 Fast Processing

Rules can be evaluated quickly, making the technique suitable for processing large numbers of emails.

3.3 Easy to Explain

The system can provide a clear reason for its decision.

For example:

"Message blocked because the sender's IP address is on the blacklist."

3.4 Effective Against Known Spam

Rule-based systems work well when the characteristics of the attack are already known.

For example:

- Known malicious IP

- Known malicious domain
 - Known phishing URL
 - Known spam keywords
 - Known malicious attachment
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4. Limitations of Rule-Based Detection

The major limitation is that it depends on **predefined rules**.

4.1 Difficulty Detecting New Spam

Attackers continuously modify their messages.

For example:

Old spam:

"WIN FREE MONEY NOW!"

Modified spam:

"You have been selected for a special financial opportunity."

If the system only looks for "FREE MONEY", the second message may bypass the filter.

4.2 Requires Regular Rule Updates

Whenever attackers develop a new technique, administrators may have to create new rules.

New Spam Attack



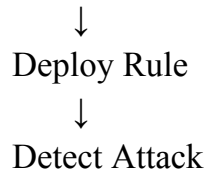
Security Team Analysis



Create New Rule



Test Rule



This makes the system less adaptive.

4.3 False Positives

A legitimate email may contain words commonly associated with spam.

For example:

"Our company is providing a free cybersecurity training program."

A simple keyword filter may incorrectly classify this email as spam.

5. Machine Learning-Based Spam Detection

Machine learning-based spam detection uses algorithms that learn patterns from previously classified emails.

The training dataset normally contains:

- Spam emails
- Legitimate emails

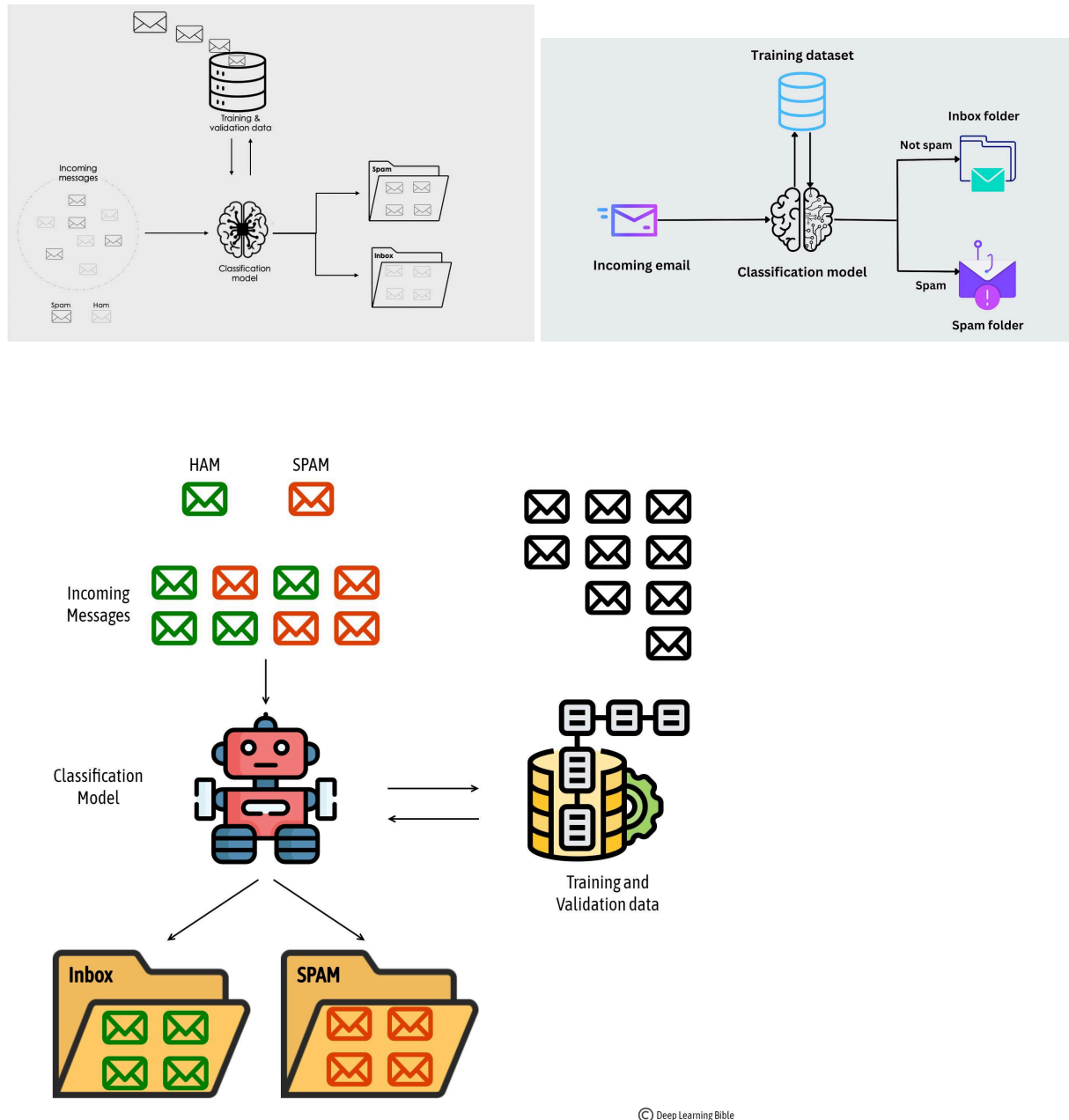
The system extracts useful features from the emails and trains a classification model.

Common algorithms include:

- Naive Bayes
- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine
- Neural Networks

Machine-learning spam filtering generally learns from examples rather than depending completely on manually written rules. Research and practical filtering systems commonly use Bayesian or other learning-based approaches alongside rule filtering. ([MDPI](#))

Picture 2 – Machine Learning Spam Detection



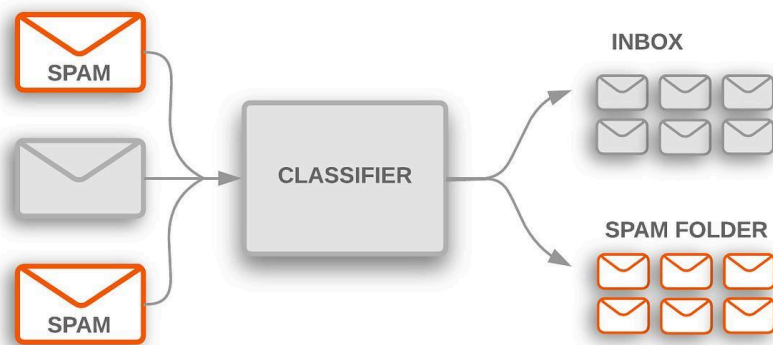
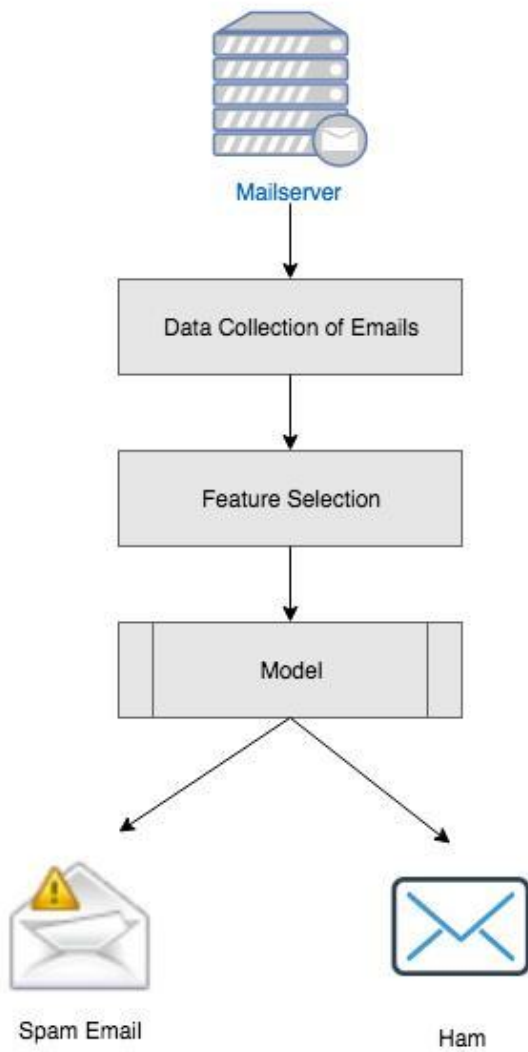
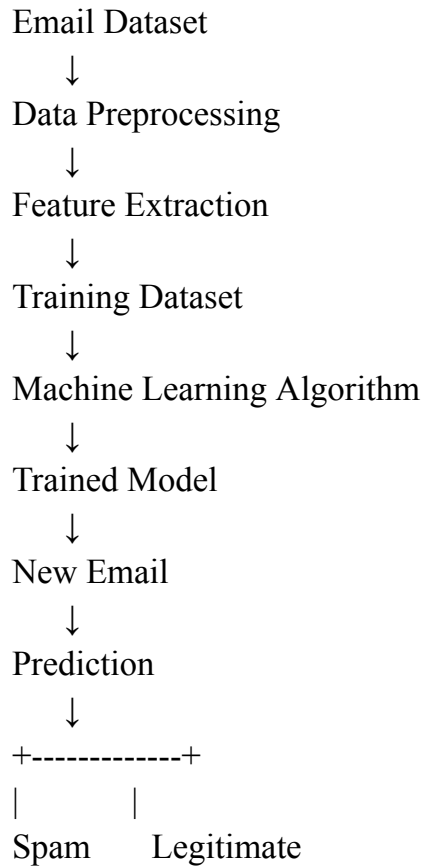


Figure 2: Machine learning-based email spam classification

6. Working of Machine Learning Spam Detection



For example, the model may analyze:

- Word frequency
- Subject characteristics
- Sender information
- URL characteristics
- Email length
- Header information
- Attachment information

Instead of checking only one keyword, the model considers multiple characteristics.

7. Advantages of Machine Learning Detection

7.1 Handles Evolving Spam Patterns

This is one of the biggest advantages of machine learning.

Attackers constantly modify:

- Words
- Sentence structures
- URLs
- Email formats
- Sender identities

A trained model can identify broader patterns rather than relying only on exact predefined keywords.

7.2 Detects Unknown Spam

Machine learning can identify suspicious emails that do not exactly match previously created rules.

This makes it more useful against changing phishing and spam campaigns.

7.3 Less Manual Rule Creation

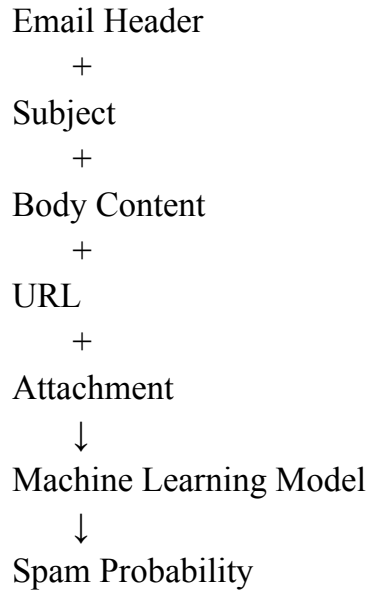
Once the model is properly trained, large numbers of emails can be automatically classified.

7.4 Multiple Features Can Be Used

The model can combine multiple signals.

Sender Reputation

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8. Limitations of Machine Learning

8.1 Requires Training Data

The model needs a sufficiently large and representative dataset.

If the training data is poor, the model's predictions may also be poor.

8.2 Model Updating

Spam patterns change over time.

Therefore, the model should periodically be retrained using newer examples.

8.3 Computational Requirements

Complex machine learning models may require more computing resources than simple rule-based filters.

8.4 False Positives

ML models can also incorrectly classify legitimate emails as spam.

Therefore, model evaluation and threshold tuning are important.

9. Direct Comparison

Feature	Rule-Based	Machine Learning-Based
Principle	Predefined rules	Learns from data
Implementation	Simple	More complex
Speed	Very high	High
Known spam	Excellent	Excellent
Unknown spam	Weak	Better
Adaptability	Low	High
Manual maintenance	High	Lower
Training data	Not required	Required
Explainability	Very high	Moderate
Evolving spam	Less effective	More effective
False positives	Can be high	Can be reduced
Scalability	Good	Excellent
Best application	Known threats	Changing threats

10. Handling Evolving Spam Patterns

Modern spam attacks change frequently.

Consider the following example.

Attack 1

Subject:

WIN FREE MONEY!!!

A rule-based filter can easily identify this.

Attack 2

Subject:

Important Financial Notification

The attacker may use a more professional-looking message.

The machine learning model can analyze additional characteristics such as:

- Sender reputation
- URL structure
- Writing patterns
- Header information
- Message similarity
- Domain characteristics

Therefore, machine learning generally provides better adaptability.

11. Minimizing False Positives

A **false positive** occurs when a legitimate email is incorrectly classified as spam.

Example

Actual Email → Legitimate

System Prediction → Spam

Result → False Positive

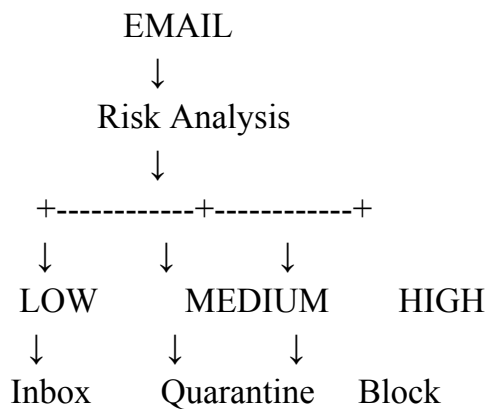
This can be dangerous in an organization because important emails may be missed.

Methods to reduce false positives:

1. Use multiple features.
 2. Analyze email headers and content together.
 3. Maintain trusted sender lists.
 4. Use confidence thresholds.
 5. Place uncertain emails in quarantine.
 6. Allow users to mark emails as "Not Spam."
 7. Regularly retrain ML models.
 8. Monitor precision and recall.
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12. Risk-Based Classification

Instead of making only a binary decision, the system can use three categories.



Low Risk

The email is delivered to the inbox.

Medium Risk

The email is sent to quarantine for further inspection.

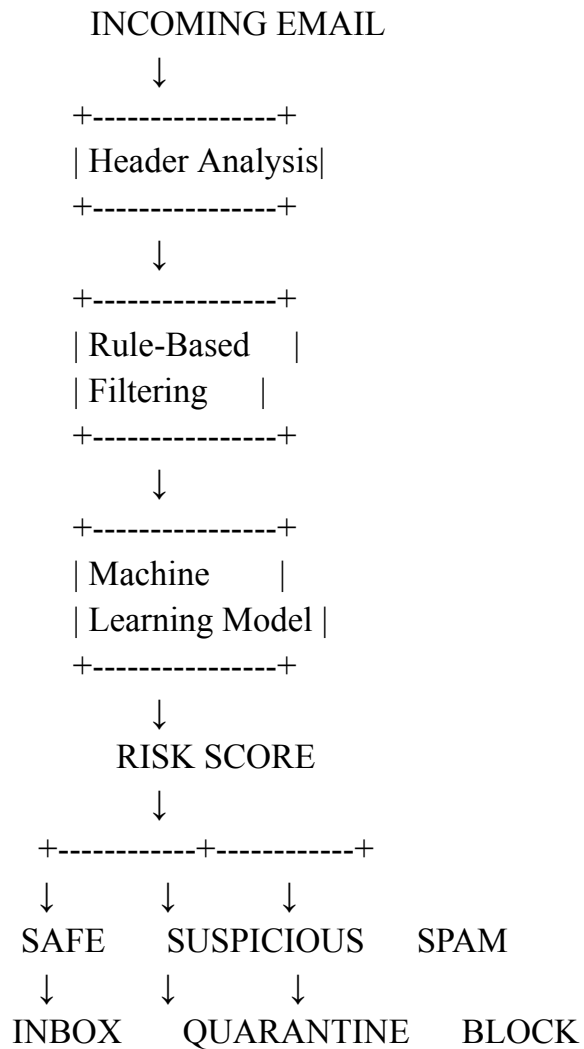
High Risk

The email can be blocked or rejected.

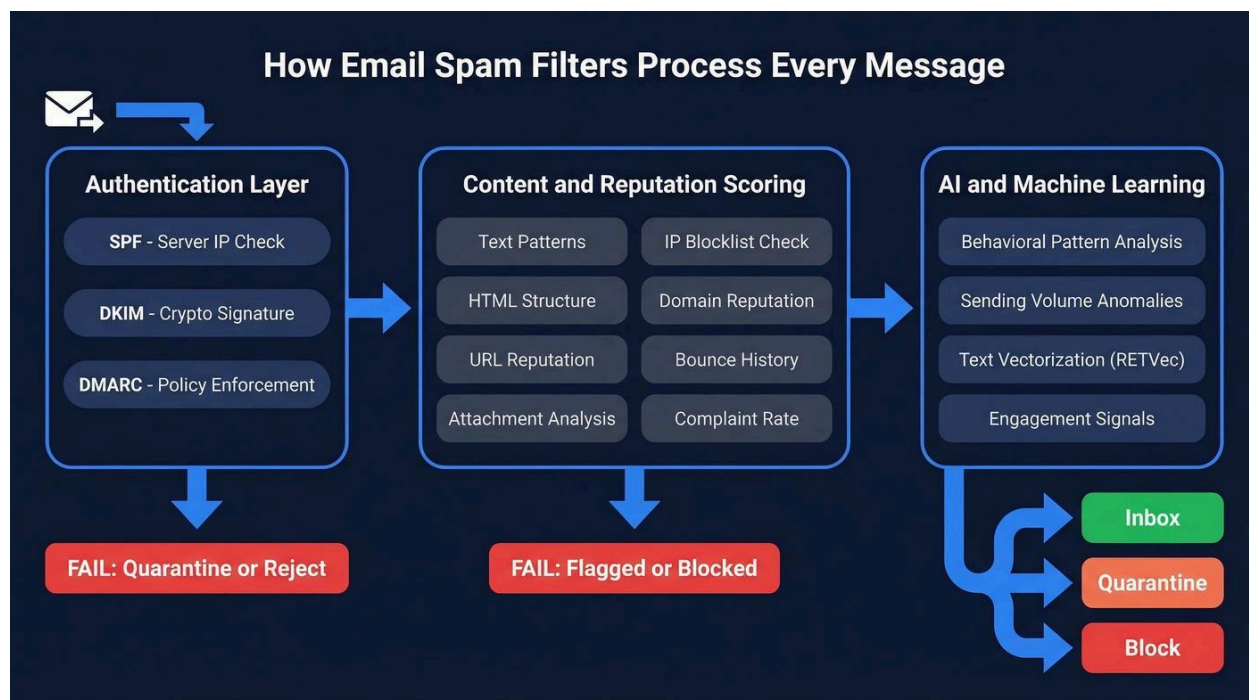
This approach helps minimize false positives because uncertain messages are not immediately deleted.

13. Hybrid Approach

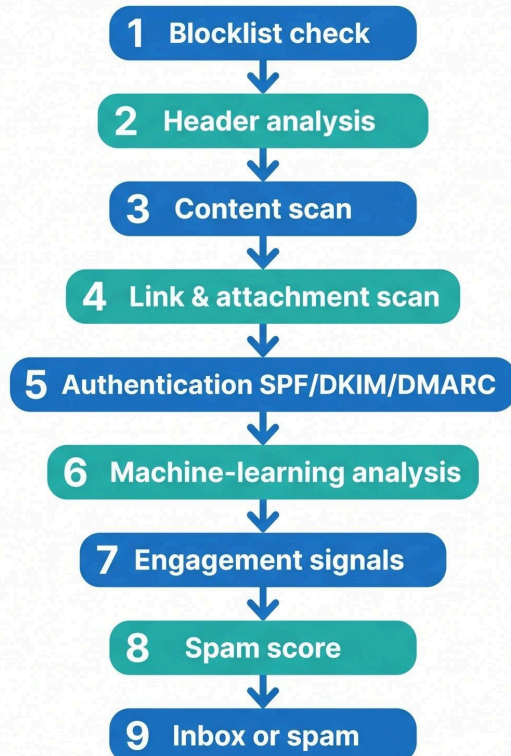
Instead of selecting only one technique, an organization can combine both.



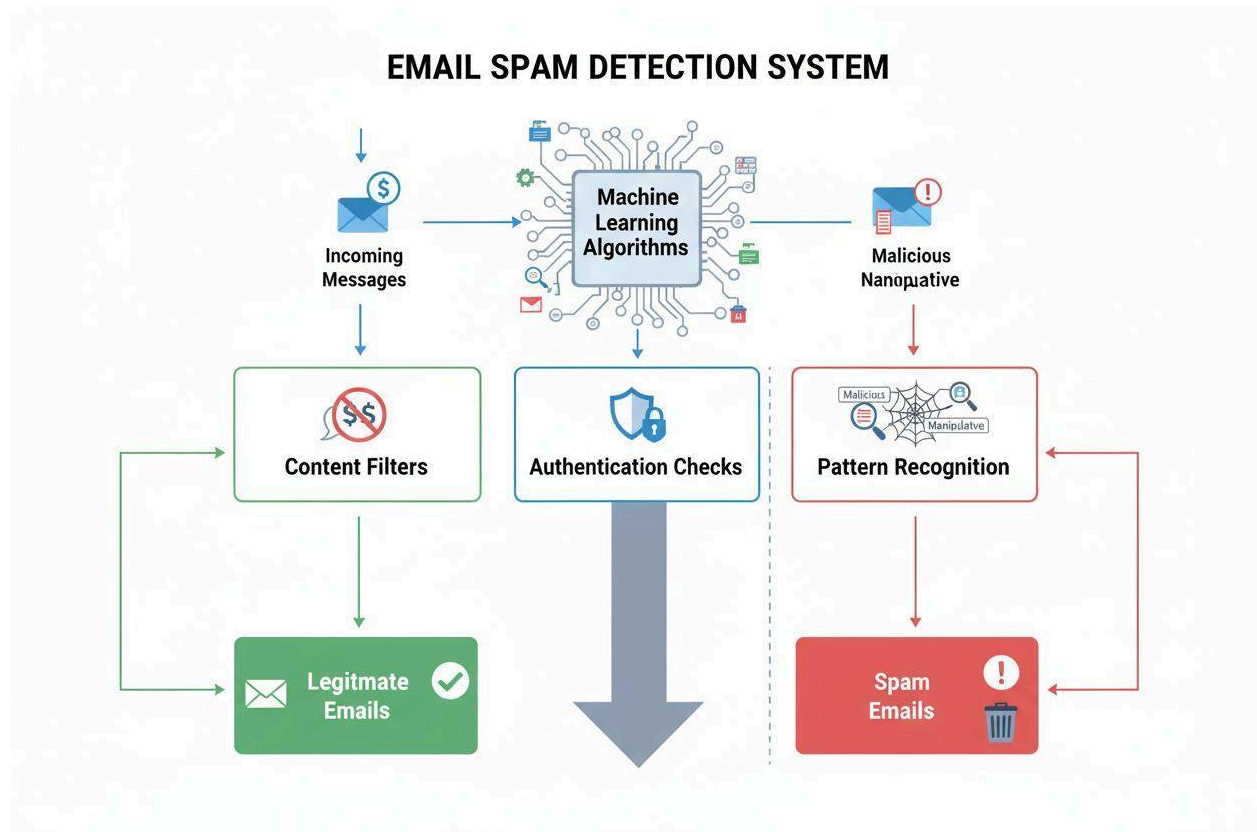
Picture 3 – Multi-Layer Spam Detection



How an email spam filter scores your message



Source: SmartReach deliverability analysis



Mailflow



Figure 3: Hybrid email spam detection architecture

Modern filtering systems can combine sender authentication, reputation checks, rules, content analysis, and learning-based scoring rather than relying on a single signal. ([MDPI](#))

14. Why Hybrid Detection is Better

The two approaches have complementary strengths.

Rule-Based Detection

Best for:

- Known malicious IP addresses
- Blacklisted domains
- Known phishing URLs
- Known malicious attachments
- Organization-specific security policies

Machine Learning

Best for:

- New spam patterns
- Changing language
- Unknown phishing emails
- Complex message patterns
- Large-scale classification

Therefore:

Rule-Based Detection + Machine Learning = Stronger and More Adaptive Email Security

17. Example

Suppose an organization receives **1,000 emails**.

Out of these:

- 700 are legitimate.
- 300 are spam.

Rule-Based System

Suppose:

- 270 spam emails are detected.
- 30 spam emails are missed.
- 40 legitimate emails are incorrectly blocked.

This means the system works well against known patterns but creates some false positives.

Machine Learning System

Suppose:

- 285 spam emails are detected.
- 15 spam emails are missed.
- 15 legitimate emails are incorrectly classified.

This demonstrates how a properly trained model can improve spam detection and reduce false positives.

18. Final Recommendation

For a modern enterprise email system, using only rule-based filtering is not sufficient because attackers continuously change their techniques.

A better solution is a **hybrid spam detection architecture** that combines:

Rule-Based Filtering

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Machine Learning

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Email Header Analysis

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Content Analysis

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Sender Reputation

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URL and Attachment Analysis

This allows known threats to be blocked quickly while machine learning identifies new and evolving spam patterns.

Conclusion

Rule-based and machine learning-based spam detection techniques both provide useful security capabilities.

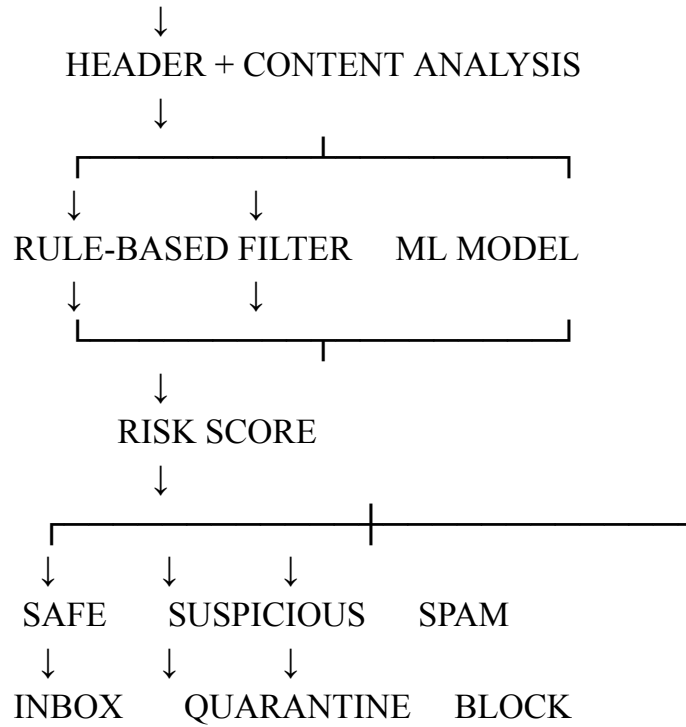
Rule-based detection is simple, fast, transparent, and highly effective against known spam patterns. However, it requires continuous rule updates and may fail when attackers modify their messages.

Machine learning-based detection is more adaptable because it learns patterns from historical data and can identify complex and evolving spam. However, it requires suitable training data, regular model updates, and careful threshold selection.

The most effective approach is therefore a **hybrid system**. Rule-based filters can handle known threats, while machine learning detects new patterns. Using quarantine for uncertain emails, trusted sender lists, user feedback, and performance metrics such as **precision, recall, and F1-score** can further reduce false positives.

Recommended Architecture

INCOMING EMAIL



This hybrid approach provides better adaptability, stronger spam detection, and improved control over false positives compared with using either rule-based or machine learning detection alone.